

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1735

Bear Island 345 kV
October 22, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1735. The Interconnection Customer proposes a 158.9 MW solar photovoltaic facility interconnecting at the Bear Island 345 kV substation in the Virginia area. The request was received on May 05, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 509 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$963,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: DUNN SIDING-BIRCHWOOD 345kV
Contingency: LEAVITT-BIRCHWOOD 345kV
Loading: 102.1% (Rating: 1,519 MVA)
DFAX: 12.9%
Pre-project loading: 100.8%
2. Facility: BIRCHWOOD-DIST 1000 138kV
Contingency: LEAVITT-BIRCHWOOD 345kV
Loading: 101.0% (Rating: 619 MVA)
DFAX: 5.5%
Pre-project loading: 99.6%
3. Facility: TRENTON DIESEL SUBSTATION-DIST 100 138kV
Contingency: LEAVITT-BIRCHWOOD 345kV
Loading: 100.8% (Rating: 633 MVA)
DFAX: 5.5%
Pre-project loading: 99.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DUNN SIDING-BIRCHWOOD 345kV	Reconductor line	\$18,887,000	2% (\$392,000)
BIRCHWOOD-DIST 1000 138kV	Reconductor line	\$7,600,000	3% (\$202,000)
TRENTON DIESEL SUBSTATION-DIST 100 138kV	Reconductor line	\$13,878,000	3% (\$369,000)
	Total allocated to project		\$963,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.